

Unit 7: Optimizing E-commerce Systems (5 Hrs.)

- Search engine
- Search Engine Optimization
- Working mechanism of Search Engines, On Page SEO, Off Page SEO, Page Ranks,
- Using Google Analytics, Social Media Analytics,
- Recommendation Systems: Collaborative, Content Based, Use of Recommendation Systems in E-commerce

Search engine

- A search engine is a software (accessible through a Web site) that helps people find things on the Web.

Components of search engine/Working mechanism of search engine

- Search engines contain three major parts: *Spider, Index and Search utility*

1. Spider :

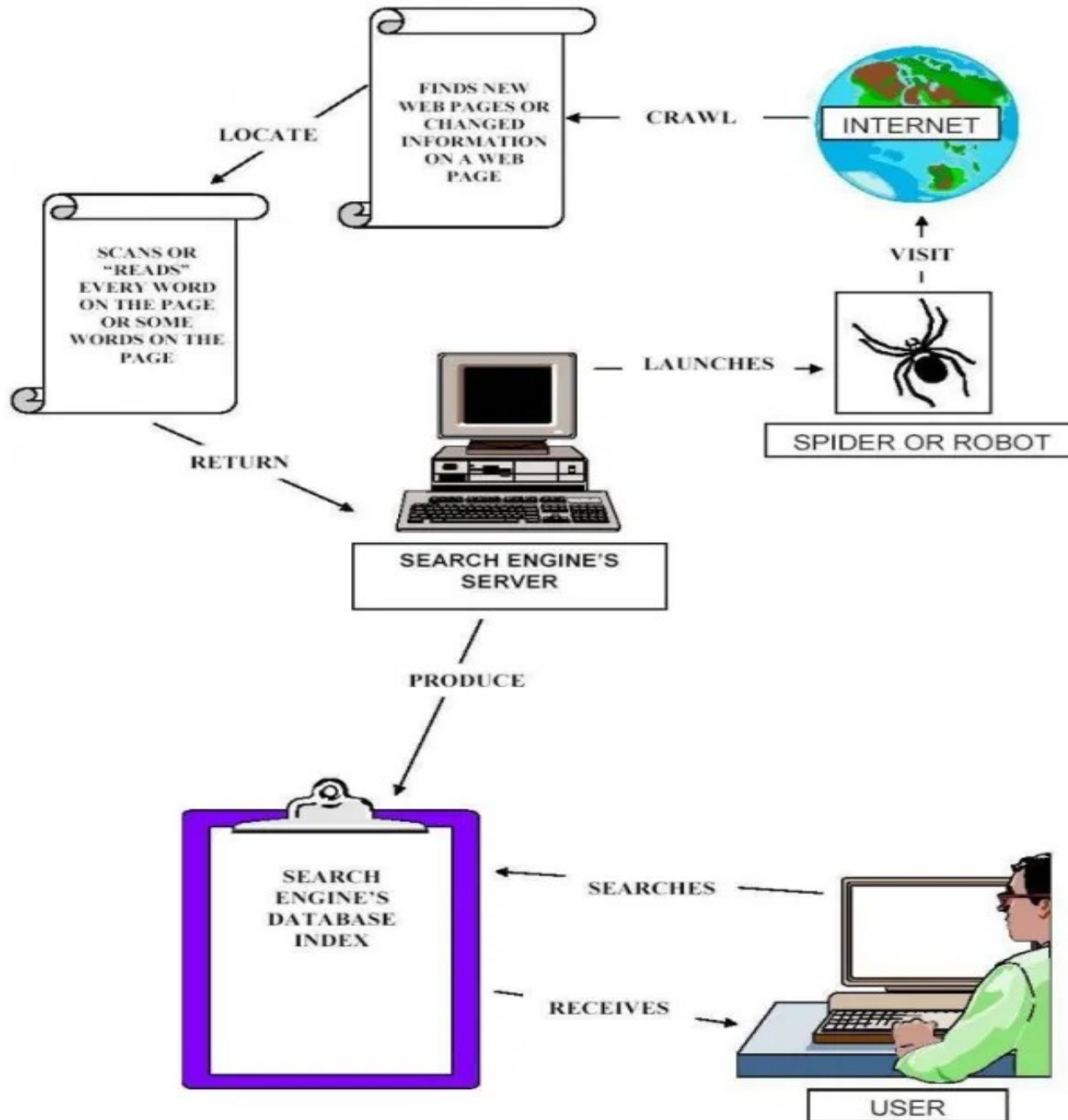
- The first part, called a spider, a crawler, or a robot (or simply bot), is a program that automatically searches the Web to find Web pages that might be interesting to people.
- When the spider finds Web pages that might interest searcher/visitor, it collects the URL of the page and information contained on the page. This information might include the page's title, keywords included in the page's text, and information about other pages on that Web site. In addition to words that appear on the Web page, Web site designers can specify additional keywords in the page that are hidden from the view of Web site visitors but that are visible to spiders. These keywords are enclosed in an HTML tag set called meta tags. The word meta is used for this tag set to indicate that the keywords describe the content of a Web page and are not themselves part of the content.

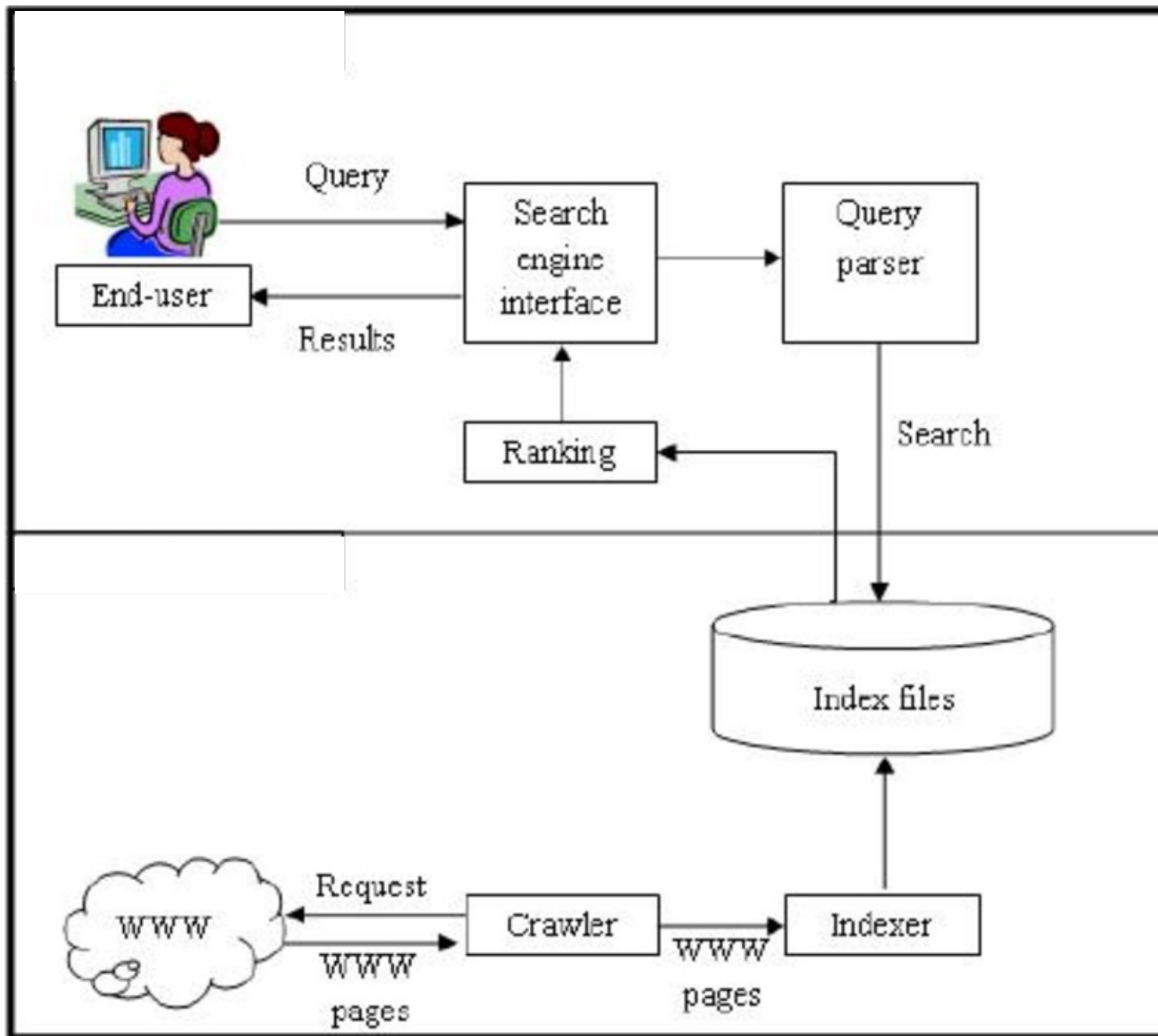
2. Index:

- The spider returns information to the second part of the search engine to be stored.
- The storage element of a search engine is called its index or database.
- The index checks to see if information about the Web page is already stored. If it is, it compares the stored information to the new information and determines whether to update the page information. The index is designed to allow fast searches of its very large amount of stored information.

3. Search Utility:

- The third part of the search engine is the search utility.
- Visitors to the search engine site provide search terms, and the search utility takes those terms and finds entries for Web pages in its index that match those search terms. The search utility is a program that creates a Web page that is a list of links to URLs that the search engine has found in its index that match the site visitor's search terms. The visitor can then click the links to visit those sites.





Search engine optimization (SEO)

- Marketers want to make sure that when a potential customer enters search terms that relate to their products or services, their companies' Web site URLs appear among the first 10 returned listings.
- The weighting (measurement) of the factors that search engines use to decide which URLs appear first on searches for a particular search term is called a *search engine ranking*.
- For example, if a site is near the top of the list of links returned for a search term "auto," that site is said to have a high search engine ranking for "auto."
- The combined art and science of having a particular URL listed near the top of search engine results is called *search engine positioning, search engine optimization, or search engine placement*.
- For sites that obtain most of their visitors from search engines, a high ranking that places their URL near the top of the list of links returned by the search engine is extremely important.

- **SEO** is the process of improving the ranking of web pages with search engines such that the website link appears on top of search results and it is easy for users to find the website.
- By carefully selecting key words used on the web pages, updating content frequently, and designing the site so it can be easily read by search engine programs, marketers can improve the impact and return on investment in their web marketing programs.

On-page SEO and Off-Page SEO

On-Page SEO:

- It is all the measures taken directly within a website to improve its position in search rankings by making it easy for search engine bots to interpret the page as well as give end-users a preview of what they're clicking through from the SERP. It also takes into consideration overall content quality, page performance, and content structure.
- On-page SEO (also known as "on-site SEO") is the practice of optimizing web page content for search engines and users. Common on-page SEO practices include optimizing title tags, content, internal links and URLs.
- On-page SEO is the practice of optimizing individual web pages in order to rank higher and earn more relevant traffic in search engines. On-page refers to both the content and HTML source code of a page that can be optimized, as opposed to off-page SEO which refers to links and other external signals.
- Some on-page SEO techniques include optimizing title tags, content, internal links and URLs.
- The most basic signal that information is relevant is when a webpage contains the same keywords as our search query. If those keywords appear on the page, or if they appear in the headings or body of the text, the information is more likely to be relevant.

Off-Page SEO:

- optimizing for signals that happen off of website
- “Off-Page SEO” refers to all of the activities that are done away from our website to raise the ranking of a page with search engines
- One of the off-page SEO is link building (having the link of our website in other websites.
- if we have got a lot of valuable links pointing to our pages, search engines will assume that our pages have great content – the type that provides value for users
- But there are other techniques also e.g. If you write a guest post for another blog or leave a comment, you’re doing off-page site promotion

Page Rank:

- PageRank(PR) is an algorithm used by Google Search Engine to rank web pages in their search engine results.
- PageRank was named after Larry Page, one of the founders of Google.
- PageRank is a way of measuring the importance of website pages.
- According to Google,
“PageRank works by counting the number and quality of links to a page to determine a rough estimate of how important the website is. The underlying assumption is that more important websites are likely to receive more links from other websites”
- Currently, PageRank is not the only algorithm used by Google to order search results, but it is the first algorithm that was used by the company, and it is the best known.



TOOLS FOR SEARCH ENGINE OPTIMIZATION

- A website is only as valuable from a business perspective as the number of people who visit.
- The first stop for most customers looking for a product or service is a search engine, and follow the top three to five listings, then view the sponsored ads.
- The higher (usually top 5 or at least the first page) our website is on the search engine pages, the more traffic it will receive.

- *How to make our website link come to Page 1 (first page) in the natural (unpaid) search listings?*
 - Every search engine is different, and none of them publish their algorithms for ranking pages.
 - Some basic ideas that work well:
 - *Metatags, titles, page contents:*
 - Search engines “crawl” web sites and identify keywords as well as title pages and then index them for use in search arguments.
 - We can develop our webpages with keywords that accurately describe the content in your metatag site “description” and “keywords” sections of your source code.
 - Experiment: use different keywords to see which work. “Vintage cars” may attract more visitors than “antique cars” or “restored cars.”

- *Identify market niches:*
 - Instead of marketing “jewelry,” we be more specific, such as “Victorian jewelry,” or “1950s jewelry” to attract small, specific groups who are intensely interested in period jewelry and closer to purchasing.
- *Offer expertise:*
 - White papers, industry analyses, FAQ pages, guides, and histories are excellent ways to build confidence on the part of users and to encourage them to see a website as the place to go for help and guidance.
- *Get linked up:*
 - Encourage other sites to link to our site.
 - build a blog that attracts people and who will share our URL with others and post links in the process.
 - Build a Facebook page for the company. Utilize Instagram or Pinterest to develop a following or fan base.

o *Buy ads:*

- Complement the natural search optimization efforts with paid search engine keywords and ads.
- Choose your keywords and purchase direct exposure on web pages.
- You can set your budget and put a ceiling on it to prevent large losses.
- See what works, and observe the number of visits to your site produced by each keyword string.

o *Local e-commerce:*

- Developing a large (national/international) market can take a long time.
- If your website is particularly attractive to local people, or involves products sold locally, use keywords that connote your location so people can find you nearby.
- Town, city, and region names in your keywords can be helpful, such as “Vermont cheese” or “San Francisco blues music.”

Recommendation system

- Recommender systems are used to predict users' interests and recommend product items that might be interesting for them.
- Also called recommender system recommendation platform or recommendation engine
- E.g. When we buy a product from an ecommerce site, then next time when we visit the website, it recommends/suggests if we are interested in buying certain product.
- Recommendation systems use two kinds of info/data to make prediction about user's preferences :
 - user rating about a product
 - search engine queries made by user
 - User's purchase histories
 - other info about the users/products themselves e.g. user profile/personal info.

Why recommender systems?/Advantages

- Increase in sales as a result of very personalized offers and an enhanced customer experience.
- User can search items very fast. Recommendations speed up searches and make it easier for users to access content they're interested in
- Increased Ad hoc purchase: Sometimes recommendation surprise users with offers they would have never searched for. And they buy the item.
- companies are able to gain and retain customers by sending out emails with links to new offers that meet the recipients' interests
- Because of recommendation, the user is more likely to buy additional products or consume more content. By knowing what a user wants, the company gains competitive advantage and the threat of losing a customer to a competitor decreases.

Types of recommendation system:

- Based on what kind of information they use, we can distinguish three algorithms used in recommender systems:
 - Content-based systems
 - Collaborative filtering systems
 - Hybrid systems
- Content-based systems
 - These systems make recommendations using a user's item and profile features.
 - They hypothesize that if a user was interested in an item in the past, they will once again be interested in it in the future.
 - Similar items are usually grouped based on their features.
 - User profiles are constructed using historical interactions or by explicitly asking users about their interests.

- There are other systems, which utilize user personal and social data.
- Limitations:
 - One issue that arises is making obvious recommendations because of excessive specialization (user A is only interested in categories B, C, and D, and the system is not able to recommend items outside those categories, even though they could be interesting to them).
 - Another common problem is that new users lack a defined profile unless they are explicitly asked for information.

Collaborative filtering

- currently one of the most frequently used approaches and usually provides better results than content-based recommendations.
- Some examples of this are found in the recommendation systems of Youtube, Netflix.
- These kinds of systems utilize user interactions to filter for items of interest.















